









Fun:

We had fun the whole season, with a core values highlight at each practice. We played Grounders and Soccer, bonded over Timbits at practice, played Guess in 10, and did hide-and-seek with timers. Even when things went wrong, we laughed, tried again, and kept going!

Impact:

We are really excited about GridLock because we can help archaeologists measure a perfect 1x1m square on uneven ground efficiently and accurately with our simple and affordable tool. There is potential for real world impact and use for GridLock which could actually be used in the field and help archaeologists spend less time gridding and more time discovering.



We used drop-on attachments for our runs, which allows us to quickly and easily put on and take off our attachments for quick changeovers.

We took the distance between missions and the attachment requirements into account when designing our mission strategy

Our Creation Process



First, we made a box robot.



Then we duplicated the bot so we could have two teams working on the robot at the same time, which is more efficient, and we can incorporate more people into the robot.



Then we worked in two teams to code as many runs and missions as possible.



We then weaved all the runs together to make a full game plan under two and a half minutes with as many points as possible.

Inclusivity and Teamwork

Our team empowers every member to contribute to the robot game at tournaments in ways that honour their individual strengths and comfort levels.







Coding

Mission Strategies

Attachments

Processes

Impact:

We are really excited about GridLock because we can help archeologists measure a perfect 1x1m square on uneven ground efficiently and accurately with our simple and affordable tool. There is potential for real world impact and use of GridLock which could actually be used in the field and help archeologists spend less time gridding and more time discovering.

Teamwork:

We collaborated and worked together as a team on the innovation and the rest of this year. Team members leveraged their strengths – coding, building, researching, meeting experts, and organizing. When parts of our Innovation prototype broke or a robot run failed, we worked together as a team to fix it.

Fun:

We had fun the whole season, with a core values highlight at each practice. We played Grounders and Soccer, bonded over Timbits at practice, played Guess in 10, and doing hide-and-seek with timers. \Even when things went wrong, we laughed, tried again, and kept going!

Inclusion:

We have 8 teammates, from grades 6-8, and we made sure everyone got involved in brainstorming, building, coding, contacting experts, and presenting. Sometimes we debated our ideas and learned that considering different views can lead t better outcomes. We even collaborated with other FLL teams!

solution

design

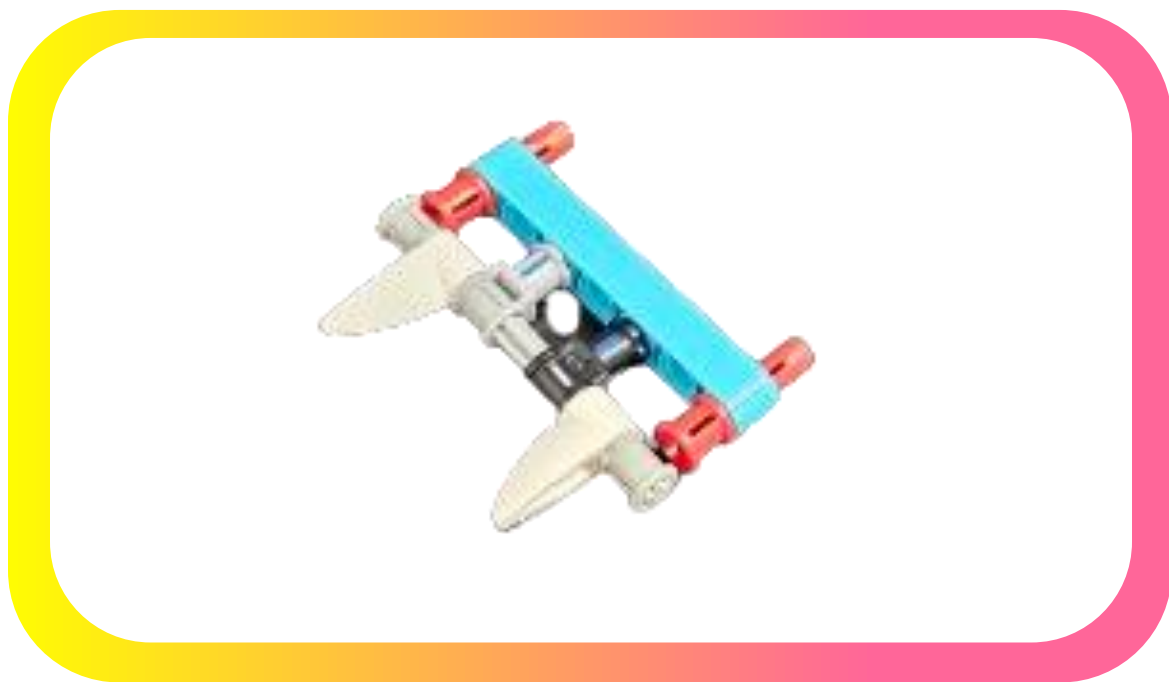


Luke LIDAR

My Favorite Core Value is...

DISCOVERY because it's really the true purpose of FLL.

Discovery is about figuring out something now and learning. We do this every practice, and it allows us to adapt to change and to innovate

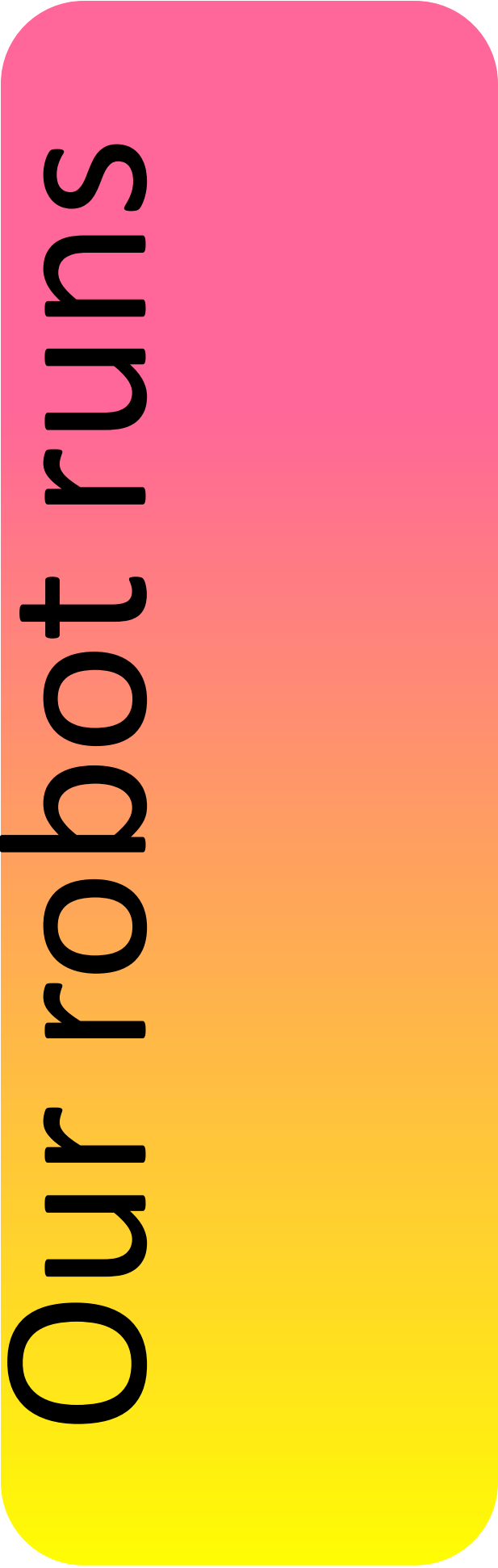


Impact:

We are really excited about GridLock because we can help archeologists measure a perfect 1x1m square on uneven ground efficiently and accurately with our simple and affordable tool. There is potential for real world impact and use of GridLock which could actually be used in the field and help archeologists spend less time gridding and more time discovering.

Teamwork:

We collaborated and worked together as a team on the innovation and the robot this year. Team members led tasks based on their strengths – coding, building, researching, meeting experts, and organizing. When parts of our Innovation prototype broke or a robot run failed, we worked together as a team to fix it.

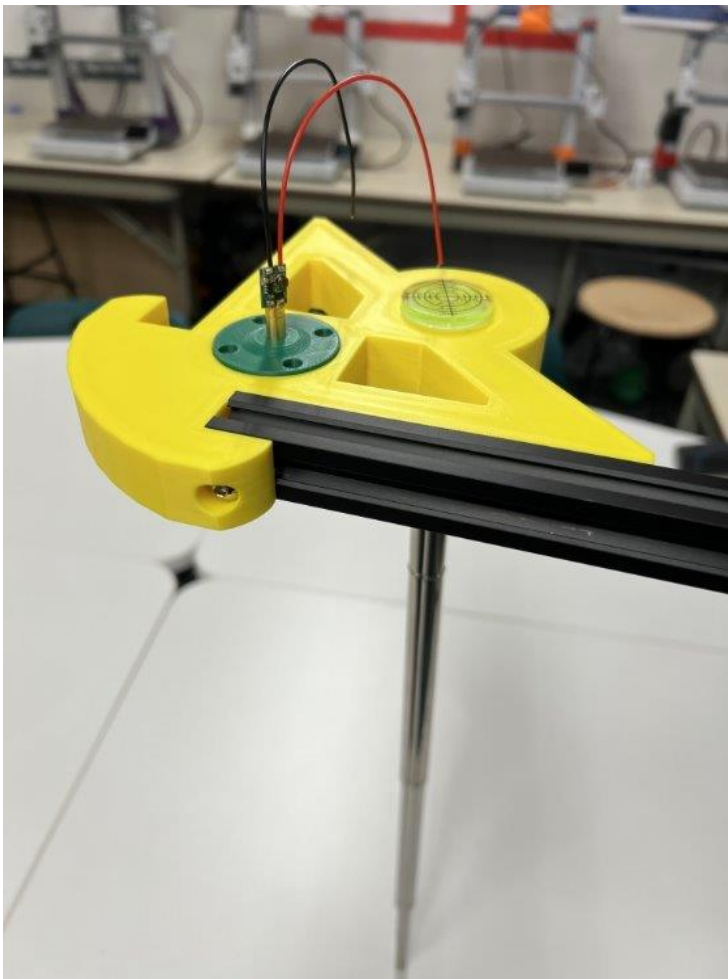


Our robot runs

















GridLock

The Next Generation Of Precision Gridding

solution

design

Problem

Testing

Experts

Financials

Materials

Impact

GridLock: Portable Precision Gridding Solution

GridLock is an innovative, portable, and affordable gridding solution designed to simplify the process of establishing precise 1×1 metre squares regardless of the ground conditions.

Key Features and Capabilities

- Lightweight aluminum frame with four independently adjustable telescoping legs
- Adaptable to sloped or uneven ground
- Omnidirectional bubble levels ensure a perfectly level top frame
- Eliminates reliance on tape measures, plumb bobs, and line levels
- Four downward lasers mark precise 1 metre x 1 metre corners
- Rapid, one-person setup requiring only two hands
- Highly scalable for larger projects—repeat, stake, and move

Grids are 1x1 meter squares used by archaeologists to accurately record where artifacts are found.

It's an essential part of the archaeology process but it's **one of the slowest, most frustrating parts of the job.**

Traditional gridding methods using tangled string, plumb bobs and tape measures are time consuming and prone to errors. Uneven terrain slows the process down even more.

If you want **professional surveying equipment**, it can cost over **\$50,000!**

Multiple resources we researched and archaeologists we interviewed talked about the challenges in the gridding process.

“Gridding is arguably the most time-consuming part of the archaeological process.”

Ramsay McKee, Cultural Resources Supervisor,
Skeetchestn Natural Resources Corporation

Gridlock will revolutionize archaeological field work

Excavation will be

FASTER, MORE PRECISE, and more EFFICIENT

SAVING ARCHAEOLOGISTS COUNTLESS HOURS

that would otherwise be spent measuring with tangled string, plumb bobs, and laying out every grid by hand

Modular design - **SINGLE TOOL TO GRID AN ENTIRE SITE**

AFFORDABLE making it accessible to all archaeologists

Less time on the GRID means more time to

DIG, EXPLORE and DISCOVER

As a team, we tested GridLock in an outdoor environment realistic to what an actual archaeological site looks like.

As a baseline we first tried to grid a square the traditional way. It was much harder than we anticipated. It took us more than an hour with 8 members just to grid one singular square!

Then we tested GridLock on a hill with rough terrain to simulate a real archaeologist's workspace. We turned on our lasers, extended the telescoping legs, and pushed down the corners until the bubble levels were perfectly aligned. Once everything was level, we placed our stakes in the ground and just like that we had a grid! It was so much faster and easier.



Revenue	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	All Future Years
Revenue YoY growth %	-	5%	5%	5%	5%	5%	5%	5%
Revenue \$'s	\$160,000	\$168,000	\$176,400	\$185,220	\$194,481	\$204,205	\$214,415	\$214,415
Expenses								
Materials	\$30,400	\$30,415	\$30,415	\$30,415	\$30,415	\$30,415	\$30,415	\$30,415
Salary	\$81,600	\$81,600	\$81,600	\$81,600	\$81,600	\$81,600	\$81,600	\$81,600
Electricity	\$3,900	\$3,900	\$3,900	\$3,900	\$5,000	\$5,000	\$5,000	\$5,000
Marketing	\$4,000	\$3,000	\$2,000	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000
Rent	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000
Total expenses	\$118,900	\$117,915	\$116,915	\$115,915	\$117,015	\$117,015	\$117,015	\$117,015
Profit	\$41,100	\$50,085	\$59,485	\$69,305	\$77,466	\$87,190	\$97,400	\$97,400

Revenue (Estimated)	
\$200.00	Selling price
10000	Number of archaeologists in North America (1,2)
4%	Market penetration in 1st year (percent of people that buy the product in the 1st year)
800	Estimated number of buyers in 1st year
\$160,000.00	Revenue in 1st year
Expenses (Estimated)	
\$10.00	Aluminum Extrusion
\$12.00	Selfie Sticks (4x)
\$3.00	Bubble levels (4x)
\$6.00	Laser modules (4x)
\$1.00	Battery Box + Batteries
\$1.00	Nuts and Bolts
\$5.00	Filament (3D printed brackets)
\$38.00	Total material cost per unit

1) CareerExplorer: The job market for archaeologists in the United States (~8,000)

2) Report of the Task Force on Discovering the Archaeologists of the Americas (referenced Canada: ~2,000)

End-User Payback Analysis					
Scenario	People Involved	Time per Task	Tasks per Day	Total Time per Day	Daily Time Difference
Current method using string, plumb bob, and measuring tape	2	10 minutes	20	400 minutes	—
GridLock	1	5 minutes	20	100 minutes	300 minutes saved (≈ 5 hours)
Pay	\$35	Dollars/Hour			
Cost of GridLock	\$200.00	Dollars			
Time saved per day	5	Hours			
Time to break even	1.14	Days			





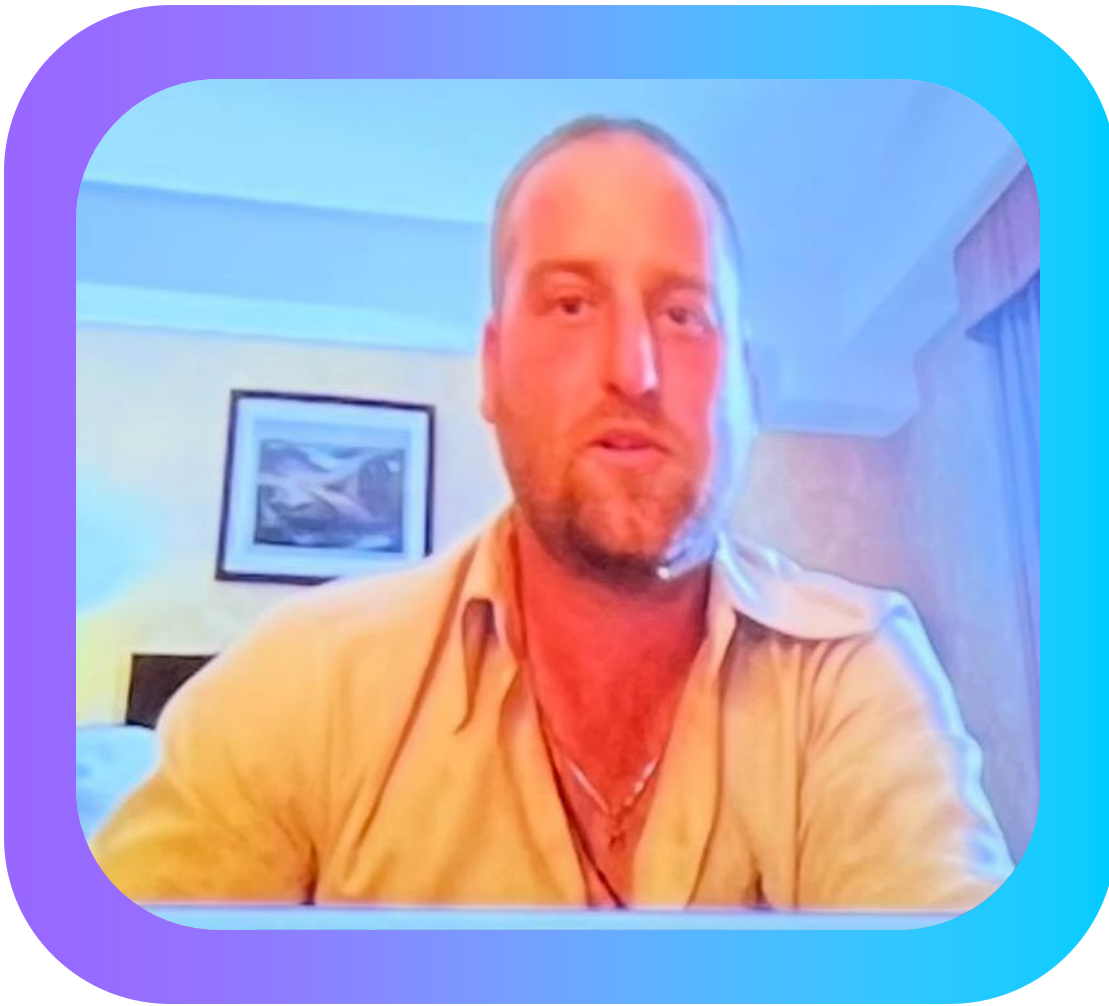
Barbara Mills

Professor of Archaeology, Arizona
State University

Curator Emeritus Archaeology, Arizona
State Museum



Lisa Sonnenburg
Cultural Resources Management
Advisor, Parks Canada



Ramsey Macfie

Archaeological Field Supervisor at
AECOM Ontario

THE FAMILY

OUR CHAIR

CORE VALUES

**HIGHLIGHT
OF THE DAY!**

Core values



Discovery:

We discovered how archaeologists uncover history and challenges they face by researching, Watching videos and meeting experts. We also learned new robot skills like how to mesh gears accurately, learning how to code bigger wheels, and designing better attachments. It's been a season of discovery for ConneCTech!

Innovation:

We innovated lots of ideas when brainstorming our innovation and evaluated our ideas for feasibility and impact. We kept improving and iterating our ideas. Innovation is process we worked on as a team through 7 iterations of GridLock.



Impact:

We are really excited about GridLock because we can help archaeologists measure a perfect 1x1m square on uneven ground efficiently and accurately with our simple and affordable tool. There is potential for real world impact and use OG GridLock which could actually be used in the field and help archaeologists spend less time gridding and more time discovering.

Teamwork:

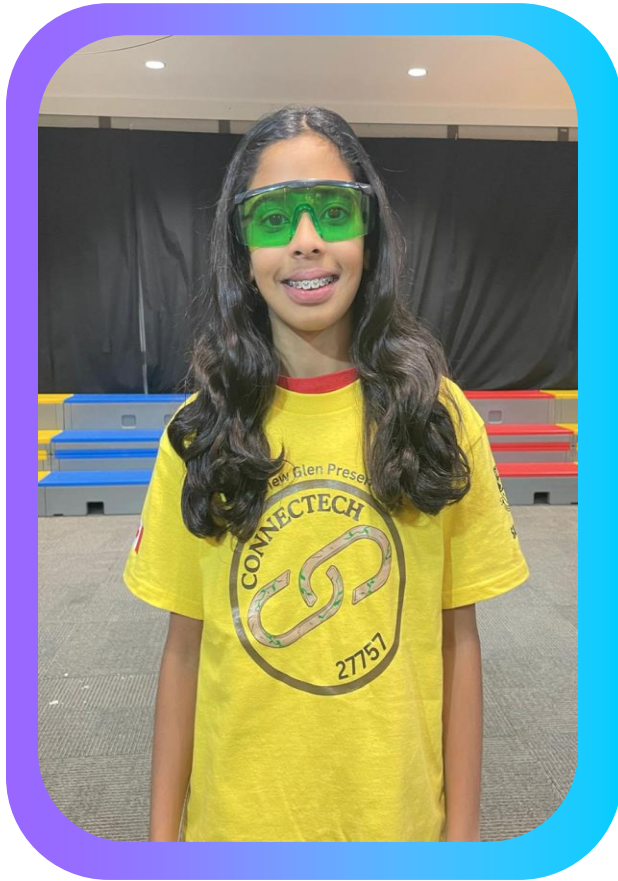
We collaborated and worked together as a team on the innovation and the robot this year. Team members led tasked based on the strengths – coding, building, researching, meeting experts, and organizing. When parts of our Innovation prototype broke or a robot run failed, we worked together as a team to fix it.

Fun:

We had fun the whole season, with a core values highlight at each practice. We played Grounders and Soccer, bonded over Timbits at practice, played Guess in 10, and doing hide-and-seek with timers. Even when things went wrong, we laughed, tried again, and kept going!

Inclusion:

We have 8 teammates, from grades 6-8, and we made sure everyone got involved in brainstorming, building, coding, contacting experts, and presenting. Sometimes we debated our ideas and learned that considering different views can lead to better outcomes. We even collaborated with other FLL teams!



Salima Sand

My Favorite Core Value is...

Impact because it goes beyond just building robots or creating ideas. The goal of FLL is to make a lasting difference not only in the challenge but in the real world. The skills we learn and the changes we create can help people and inspire others.



Grace Grid

**My Favorite Core Value
is...**

INNOVATION because innovation helps you, not just now but in the future. It helps you become more creative by making you think more innovatively, which will lead to you becoming a better version of yourself.



Sarina Stone

My Favorite Core Value is...

INCLUSION because it gives everyone a chance to take part in things they might not usually do, and to help people feel comfortable to share ideas. Inclusion means making sure everyone feels valued and has a voice.



Luke LIDAR

My Favorite Core Value is...

DISCOVERY because it's really the true purpose of FLL.

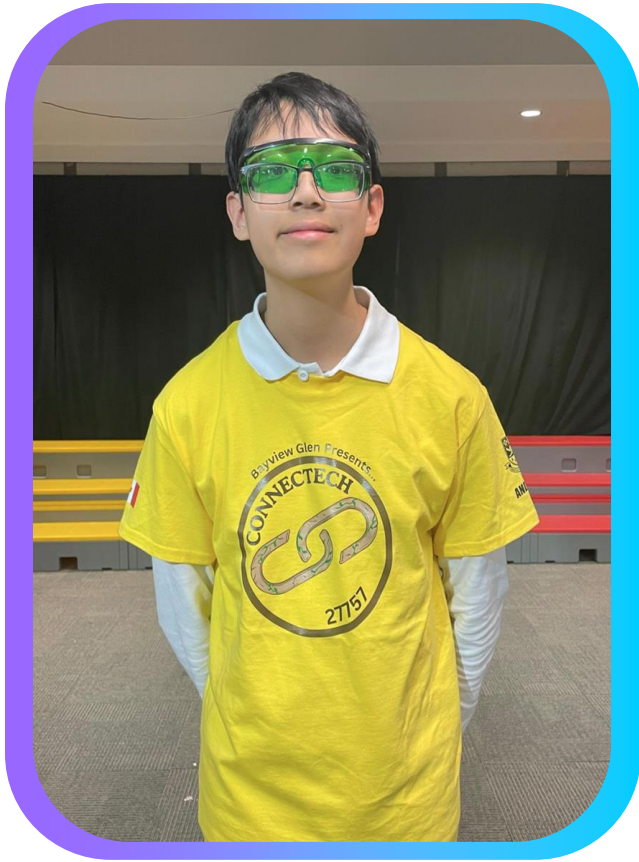
Discovery is about figuring out something now and learning. We do this every practice, and it allows us to adapt to change and to innovate



Ivan Identify

My Favorite Core Value is.....

FUN because having fun is very important for good mental health. You can achieve much more if you are having fun while working and having fun is a core part of what makes the FLL experience great.



Andrew Artifact

My Favorite Core Value is...

IMPACT because the main purpose is to find a solution that will solve a problem in the world. Some teams have actually gone the distance to advance their innovation projects beyond the competition, which shows that this robotics foundation can make a real difference in the world.



Eric Excavate

My Favorite Core Value is...

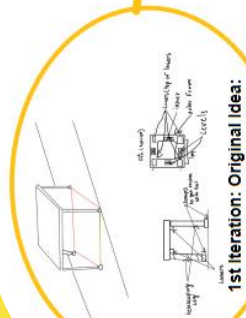
INNOVATION: because it's about creativity and finding new ways to solve problems, something all FLL teams need. Innovation helps us think outside the box, and coming up with new ideas that can make a difference



Naya Natural

My Favorite Core Value is...

TEAMWORK because you can achieve so much more when you are working with a team. Teamwork help everyone share their ideas, feel included and make tasks easier and more enjoyable. It also teaches people communication, trust and respect which are very important life skills.



1st Iteration: Original Idea:



2nd Iteration: Wooden Frame:



3rd Iteration: PVC Pipes:



4th Iteration: Aluminum
Extrusion:



5th Iteration: Second Battery

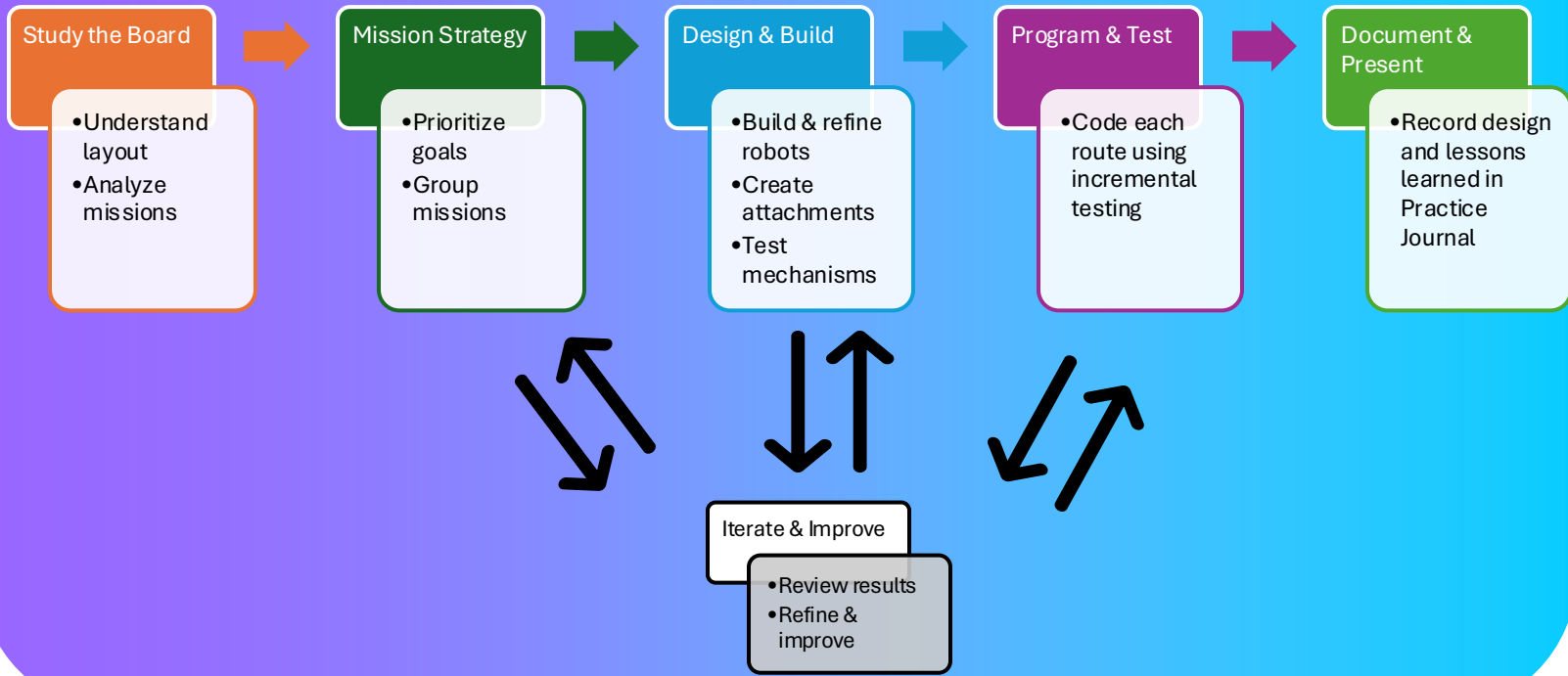


6th Iteration: Improved Threading:



7th Iteration: Improved Threading (Again):

Iteration Process



Our Creation Process



First, we made a box robot.



Then we duplicated the bot so we could have two teams working on the robot at the same time, which is more efficient, and we can incorporate more people into the process.



Then we worked in two teams to code as many runs and missions as possible.



We then weaved all the runs together to make a full game plan under two minutes with as many points as possible.

Timeline

Sept

- Understand team missions, team structure, and safety practices
- Get all team members familiar with Block coding
- Create mission strategy
- Build Waddles and Twaddles base robot frames

Oct

- Further build of robots
- Build mission-specific attachments and test mechanical concepts, such as alignment features
- Start coding runs 1 and 2

Nov

- Start coding runs 3 to 5
- Fine-tuning all the runs
- Practice full run sequences under time pressure and perform final quality checks
- Prepare documents and compile operator list for tournament

Understanding Missions

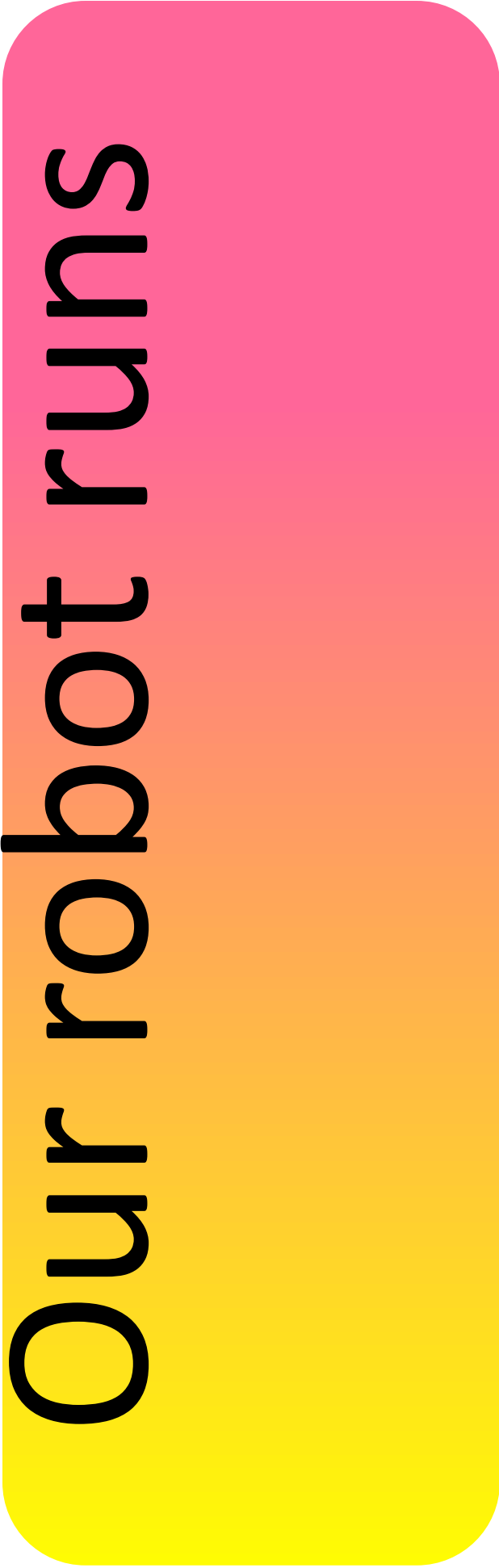
- Maximum points: 545
- Number of missions: 15
- Number of home areas: 2
- Only 3 line-following paths
- Several missions share similar movements, including horizontal pushing actions

Defining Goals

- Point target: 400
- Minimize number of attachments and simplify their designs
- Build a compact, agile, boxed-shaped robot to improve maneuverability and alignment

Our robot runs





Our robot runs

Coding Approach

The team chose to use block coding for our robot because it offers several advantages:

- All team members are already comfortable with block coding
- User-friendly interface
- Minimizes syntax errors that are more common in text-based languages
- Can quickly test, modify, and refine code, which speeds up mission development and improves run consistency

Run 1: **Run 2:** **Run 3:**
Run 4: **Run 5:**

M01: Surface Brushing; 20 pts
M02: Map Reveal; 20 pts
M03: Mineshaft Explorer; 30 pts
Return to West Home
Total: 70 pts

M12: Salvage Operation; 30 pts
M10: Tip the Scales; 20 pts
M11: Angler Artifacts; 40 pts
Return to East Home
Total: 90 pts

M08: Silo; 30 pts
M06: Forge; 30 pts
M05: Who Lived Here?; 30 pts
M07: Heavy Lifting; 30 pts
Return to East Home
Total: 120 pts

M15: Site Marking; 10 pts
M09: What's On Sale?; 20 pts
Return to East Home
Total: 30 pts

M13: Statue Rebuild; 30 pts
M14: Forum; 15 pts
Total: 45 pts

Code Examples

Example of Turn with Heading Correction

```
define R Spin w/ Heading Correction degrees degrees speed speed

set movement motors to A+B
set movement acceleration to slow
set yaw angle to 0

wait until is yaw angle in between -1 and 1 ?

repeat until degrees - 1 < yaw angle
  if degrees - 30 > yaw angle then
    set movement speed to speed %
  else
    set movement speed to .25 * speed %
  start moving right: 100

stop moving
wait .1 seconds

if degrees < yaw angle then
  repeat until yaw angle = degrees
    set movement speed to 0.1 * speed %
    start moving left: -100
else
  stop moving

stop moving
set yaw angle to 0
set movement acceleration to medium
```

Example of Straight Drive with Proportional Heading Correction and Acceleration/Deceleration

```
define Accel Rev CM CM Speed Speed

set movement motors to B+A
A set relative position to 0
B set relative position to 0
A set motors to hold position at stop
B set motors to hold position at stop

wait until is A relative position in between -2 and 2 ?
wait until is B relative position in between -2 and 2 ?

set yaw angle to 0

wait until is yaw angle in between -1 and 1 ?

set rev Traveled Distance to 0
set rev Target Distance to CM * 360 / 19.6
set rev Accel Distance to 10 * 360 / 19.6
set rev Decel Distance to 15 * 360 / 19.6

repeat until rev Traveled Distance > rev Target Distance
  set rev Traveled Distance to abs of A relative position / 2 + abs of B relative position / 2
  if rev Traveled Distance < rev Accel Distance then
    set movement speed to abs of rev Traveled Distance + 30 / rev Accel Distance * Speed %
    start moving -0.5 * yaw angle
  else
    if rev Traveled Distance > rev Target Distance - rev Decel Distance then
      set movement speed to rev Target Distance - rev Traveled Distance + 15 / rev Decel Distance * Speed %
      start moving -0.5 * yaw angle
    else
      set movement speed to Speed %
      start moving -0.5 * yaw angle

stop moving
wait 0.01 seconds
```

Inclusivity and Teamwork

Our team empowers every member to contribute to the robot game at tournaments in ways that honour their individual strengths and comfort levels.



Our Workflow and Quality Process

Code Organization

We organized the program to make it simple for anyone to run and read. We gathered all the code and added notes, so everyone knew where it should be. We also created all our codes on Teams so we wouldn't lose our code, and anyone can edit.

Testing

We tested each mission numerous times, then we combined the missions into runs. Next, we test each run dozens of times. After that, we combined the runs into a full robot game. Finally, we run through all the runs countless times to make sure everyone is very familiar with the robot, and then everyone is ready for competition.

Robot Run Preparation Checklist

The team developed a checklist based on testing to help ensure consistent, accurate, and repeatable runs during competition.

- Verify that all field elements and mission models are set up correctly on the game board
- Check batteries are fully charged and securely attached
- Inspect the robot and attachments to confirm all components are secure and properly fitted
- Clean all wheels with lint roller
- Place all attachments in clearly marked areas for quick and efficient changeovers
- Confirm that everyone understands how to align the robot and change attachments for all the runs

Our Workflow and Quality Process (cont'd)

Progress Tracker

A tracking table was created to monitor the tasks for each mission and run. This tool kept the team aligned and provided a clear view on what was completed and what still needed attention.

	A	B	C	D	E	F	G	H	I	J	K
1	Mission #	Attachment #	Run #	Coded	Consistent	Coded #2	Consistent #2	Coded #3	Consistent #3	Points last	
2	1: Surface Brushing		3	1	☐	☐	☐	☐	N/A	N/A	0
3	2: Map Reveal		3	1	☐	☐	☐	☐	X	X	10
4	3: Minestaff Explorer		3	1	☐	N/A	N/A	N/A	N/A	N/A	0
5	4: Careful Recovery		3	1	☐	☐	☐	☐	N/A	N/A	0
6	5: Who Lived Here?		2,4	☒	☐	N/A	N/A	N/A	N/A	N/A	0
7	6: Forge		2,4	☒	☐	N/A	N/A	N/A	N/A	N/A	0
8	7: Heavy Lifting		2	☐	☐	N/A	N/A	N/A	N/A	N/A	0
9	8: Silo		2,2	☒	☒	N/A	N/A	N/A	N/A	N/A	0
10	9: What's on Sale?		4	☐	☐	☐	☐	N/A	N/A	N/A	0
11	10: Tip the Scales		4	☒	☒	☐	☐	N/A	N/A	N/A	0
12	11: Angler Artifacts		1,2,3	☒	☐	N/A	N/A	N/A	N/A	N/A	0
13	12: Salvage Operation			☒	☒	☒	☒	N/A	N/A	N/A	0
14	13: Statue Rebuild			☐	☐	N/A	N/A	N/A	N/A	N/A	0
15	14: Forum (see below)		☒								15
16	15: Site Marking		X	☒	☒	☐	☐	☐	☐		0
17											
18	Forum:										
19	Brush		☒								
20	Topocell		☐								
21	One		☐								
22	Milestone		X								
23	Scale Plan		☐	maybe		Attachment:		1.			
24	Minecart					Run 1					
25	Artifact		☐			Run 2					
26			50 (max)			Run 3					
27	Equipment Inspection					Run 4					
28	Precision Tokens					Run 5					
29											
30											
31											
32											

Lessons Learned

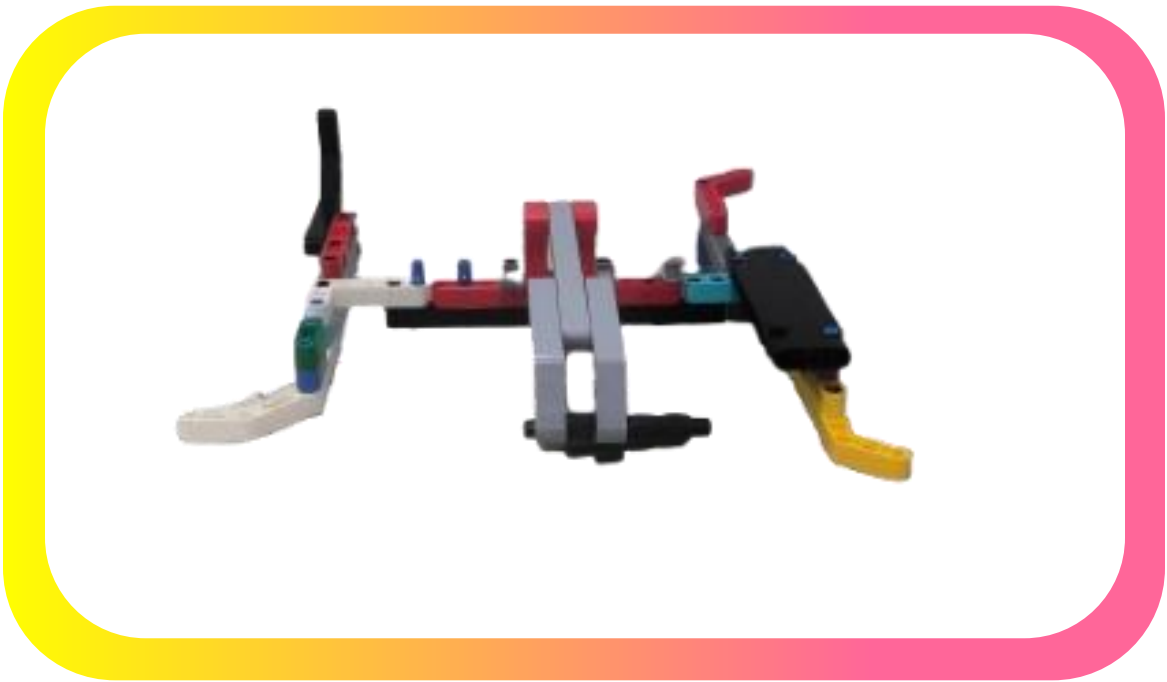
Key takeaways after each practice were documented in the Practice Journal and shared to help both current team members improve and future ConnectTech teams build on our experience.

Coding

Mission Strategies

Attachments

Processes



NEED TO GET NEW PICTURE OF
ROBOT: OLD ONE HAS GLARE

The Snake

Does M02 Map Reveal in Run 1.

The Scorpion

Does the M01 Surface Brushing and the M03 Mineshaft Explorer in Run 1.

Lionel Meshi

Does the M12 Salvage Operation, M10 Tip the Scales, and M11 Angler Artifacts in Run 2.

The Thwacker

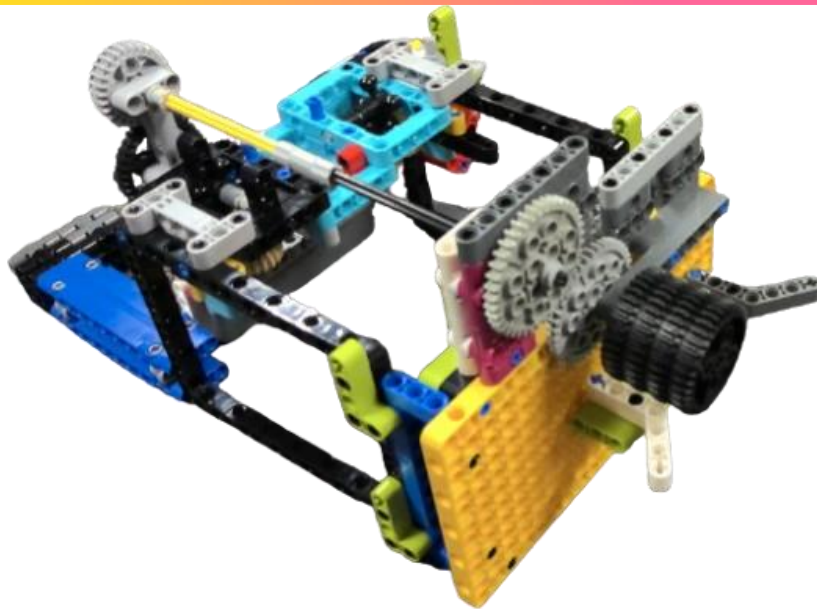
Does the M08 Silo, M06 Forge, M05 Who Lived Here?, and M07 Heavy Lifting in Run 3

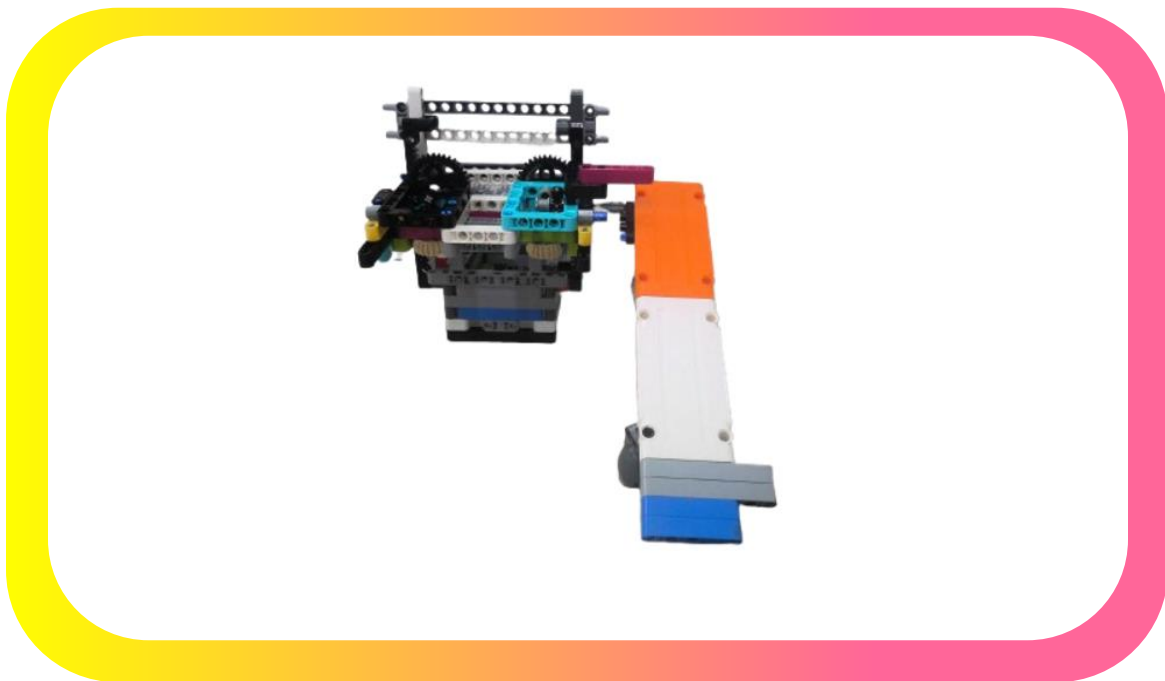
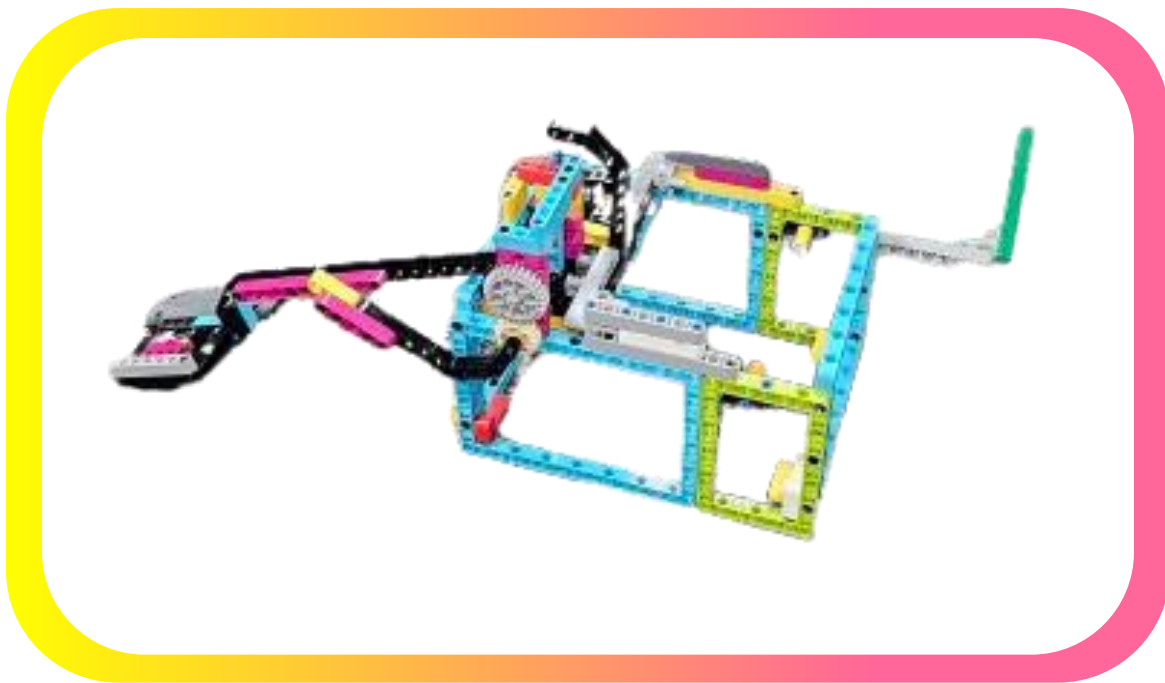
The Fangs

Does the M15 Site Markings and the M09
What's On Sale? in Run 4.

The Smasher

Does the M13 Statue Rebuild and M14
Forum in Run 5.





We used drop-on attachments for our runs, which allows us to quickly and easily put on and take off our attachments for quick changeovers.

We took the distance between missions and the attachment requirements into account when designing our mission strategy